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Anonymous authors

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Successful multi-agent interaction requires an agent to infer what its partners know and want, then choose actions in light of those inferences. Terminal success alone obscures which step fails. We study these coupled capabilities—interactive-state inference and interaction planning—in BENAC-I, a finite negotiation game with private preferences, mixed incentives, and costly investigation. Its explicit information structure supports a verified reference policy from which we compute beliefs over partner preferences and counterfactual action values. We use this oracle to construct diagnostic cases from observed partner choices and evaluate inference and planning at the resulting decisions. The diagnostic reveals failures at both stages: agents can misread behavioral evidence and make suboptimal decisions even when given the correct partner-state judgment; correcting that judgment can therefore fail to improve the chosen action. We then construct *game slices*, contiguous stretches of one or more decisions, to supervise inference and planning directly. We compare slice training with complete-game self-play on held-out negotiation structures and test transfer to CalBench and ShapeFactory.

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Large language models (LLMs) increasingly operate in multi-agent systems that distribute work and coordinate decisions through interaction (Zhu et al., 2025; Chen et al., 2026). Yet interaction can fail at several distinct points. In tasks with distributed information, agents leave partners’ private evidence unexplored (Li et al., 2026a); in distributed computation, they exchange messages without turning them into effective joint reasoning (Zhang et al., 2026b); and in negotiation, sharing the relevant information does not by itself ensure a coordinated plan (Yao et al., 2026). A broader analysis likewise identifies inter-agent misalignment as a distinct failure class (Cemri et al., 2025). These findings raise a basic question: what capabilities are missing for individually capable agents to continuously fail in interaction?

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Interactive decision theory gives this question a precise form. An interactive POMDP represents uncertainty over both the physical environment and models of other agents, and selects actions on the basis of beliefs about both (Gmytrasiewicz & Doshi, 2005). Opponent-modeling work similarly links inferences about others’ goals or policies to the planning of a response (Carmel & Markovitch, 1996; Huang et al., 2024). We therefore distinguish two capabilities: **interactive-state inference**, which estimates other agents’ task-relevant latent states from their behavior, and **interaction planning**, which uses those estimates to choose an action. We refer to them as *inference* and *planning* below. These two capabilities form a feedback loop in interaction: inference informs the next action, while that action changes how others respond and what can be inferred thereafter. Evidence from LLM agents suggests that representing or predicting another agent’s state need not translate into an effective response (Riemer et al., 2025; Sobotka et al., 2026). This gap motivates examining inference and planning both separately and together.

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Realistic multi-agent interactions are difficult to diagnose because information, objectives, and decisions are distributed across agents and time. Coordination around private calendars, negotiation over asymmetric goals, and commerce on behalf of separate principals illustrate this complexity (Zou et al., 2026; O’Neill et al., 2026; Li et al., 2026b). Games offer a controlled setting: their information structure, utilities, legal actions, and strategic consequences can be specified and verified. Recent work has used multi-agent games and self-play to train strategic or transferable reasoning

in LLMs (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b). Yet a game-level outcome alone does not reveal whether an agent misread a partner or failed to act on a correct inference. Compact games can simplify away private information or mixed incentives, while richer social games may entangle deduction, deception, commitment, and execution in a single trajectory (Light et al., 2023). To retain both capabilities while making them separately measurable, we extend the commitment-based multi-party negotiation game (Benac et al., 2026) with private preferences and costly investigation action. Incomplete information makes relevant partner states latent, creating an inference problem; mixed incentives make those states consequential for planning.

Using this controlled setting, we make the two capabilities separately testable and trainable in LLM-agent interaction. We first construct diagnostic decision problems showing that inference and planning are separated capabilities that will affect the interaction, and they take effect dependently in an long horizon case (Section 3). We then build *game slices* from deliberately structured small games: contiguous stretches of interaction containing one or more decisions, for which the reference policy provides verifiable partner-state labels and counterfactual action values. These slices provide targeted training signals for decisions that terminal game reward alone cannot disentangle. Finally, we compare game-slice training with complete-game self-play in terms of training-token efficiency, evaluate generalization to held-out game structures, and test transfer to more complex multi-agent coordination environment (Section 4).

2 MOTIVATION AND METHODS

2.1 FROM INTERACTIVE UNCERTAINTY TO TWO CAPABILITIES

Consider three coding agents working on the same Python backend. In an Anthropic experiment motivated by behavior observed in deployment, each was instructed to migrate it to a different language. The agents mistook the resulting interference for deliberate obstruction and escalated to repeatedly killing competing processes and, in some runs, revoking one another’s access (Anthropic, 2026). They depended on a usable shared backend, yet pursued incompatible local objectives without knowing the others’ instructions. Such coordination is naturally an interactive partially observable decision problem. For agent i , an I-POMDP is $\mathcal{I}_i = \langle IS_i, A, T_i, \Omega_i, O_i, R_i \rangle$, $IS_i = S \times \prod_{j \neq i} M_j$ where S is the physical state, M_j models agent j , A is the joint action space, and T_i, O_i, R_i specify transition, observation, and reward (Gmytrasiewicz & Doshi, 2005). Uncertainty therefore covers not only the world but also others’ information, preferences, and policies. This view underlies opponent modelling and belief-space coordination (Carmel & Markovitch, 1996; Huang et al., 2024; Goel et al., 2026); the observation–belief–action links nevertheless remain fragile for LLM agents (Sobotka et al., 2026; Jha et al., 2026). Let $b_t^i \in \Delta(IS_i)$ be agent i ’s belief after public history h_t . An observation updates it as $b_{t+1}^i = \tau_i(b_t^i, a_t^i, o_{t+1}^i)$, while an action is selected by belief-conditioned planning, $a_t^{i,*} \in \arg \max_{a \in A_i} \mathbb{E} \left[\sum_{k=t}^T \gamma^{k-t} R_i(x_k, a_k) \mid b_t^i, a_t^i = a, \pi_{-i} \right]$. We call these computations *inference*, which maintains the part of b_t^i predictive of others, and *planning*, which evaluates actions under that belief. They form a loop: inference informs an action, the action determines the next observation, and that observation informs the next inference. For this distinction to be experimentally useful, the hidden state must be both inferable and consequential: interaction must provide evidence that supports a reasonable belief about another agent, and different beliefs must affect the relative value of available actions. We therefore need decision points at which evidence, private information, and action consequences are all specified well enough to evaluate the belief and the choice separately. In the game below, we operationalize inference with an inference target and planning with counterfactual action values.

2.2 NEGOTIATION WITH PRIVATE PREFERENCES AND MIXED INCENTIVES

To make inference and planning separately observable while retaining exact supervision, we extend the commitment-based negotiation game of Benac et al. (2026) with private preferences and costly information gathering. An instance of this game is

$$\mathcal{E} = \langle \mathcal{N}, \mathbf{K}, \mathcal{G}, \Gamma, \delta, \rho, C_0, \mathcal{L}, \{\Theta_i\}_{i \in \mathcal{N}}, \mu_0, \mathbf{q}_0 \rangle. \quad (1)$$

Here $\mathcal{N}$ is the player set and $\mathbf{K} = (K_i)_{i \in \mathcal{N}}$ gives each player’s number of binary commitment coordinates. Each public goal $g \in \mathcal{G}$ has a nonempty requirement set $\Gamma_g \subseteq \{(i, k) : i \in \mathcal{N}, 0 \leq$

$k < K_i\}$ and a scoring mode $\delta_g \in \{\text{binary}, \text{linear}\}$. The public schedule $\rho = (\rho_t)_{t=0}^{T-1}$ specifies the proposer at each of T proposal opportunities, and $\mathcal{L}$ gives the legal actions at each information set. The commitment state is $C = (c_{ik})$, with $c_{ik} \in \{0, 1\}$; initially $C_0 = 0$, and a commitment cannot be withdrawn. Each player privately observes its preference vector $v_i \in \Theta_i \subseteq \{-1, 0, +1\}^{|\mathcal{G}|}$. The joint type $v = (v_i)_{i \in \mathcal{N}}$ is drawn from the common prior $\mu_0 \in \Delta(\prod_{i \in \mathcal{N}} \Theta_i)$; entries encode AVOID, NEUTRAL, and WANT, respectively. $\mathbf{q}_0 = (q_{i,0})_{i \in \mathcal{N}}$ gives the initial investigation budgets. For any commitment state C , goal satisfaction and terminal utility are

$$S_g(C) = \begin{cases} \mathbf{1}[c_{ik} = 1, \forall (i, k) \in \Gamma_g], & \delta_g = \text{binary}, \\ \frac{1}{|\Gamma_g|} \sum_{(i,k) \in \Gamma_g} c_{ik}, & \delta_g = \text{linear}, \end{cases} \quad U_i(C; v_i) = \sum_{g \in \mathcal{G}} v_{ig} S_g(C). \quad (2)$$

At initialization, the goals, prior, legal-action rules, and proposer schedule are public, while each player sees only its own sampled preferences; every player starts with one investigation opportunity ($q_{i,0} = 1$). Play follows ρ . At each proposal opportunity, the scheduled player may PASS, make a bilateral OFFER to a chosen partner, or, if $q_{i,t} > 0$, use INVESTIGATE(j, g) to learn v_{jg} . Passing leaves commitments unchanged. Investigation also leaves them unchanged, uses that proposal opportunity and one unit of budget, and reveals the answer only to the investigator; the query target is public. An offer specifies new commitments for the proposer, the partner, or both. The partner immediately chooses ACCEPT or REJECT: acceptance makes all proposed additions binding and irreversible, whereas rejection changes no commitments. The response does not use another proposal opportunity. Play then proceeds to the next scheduled proposer and ends after the final opportunity, when each player receives $U_i(C_T; v_i)$. To improve its utility, a player infers partners' preferences from offers, responses, and any private investigation answers, then plans whom to approach, what to offer, and whether to accept, accounting for goals on which preferences align or conflict.

2.3 GAME SLICES AND ORACLE SOLVER

A *game slice* is a consecutive segment of play that begins at a chosen decision and may include later decisions. For example, it can contain an offer, the partner's response, and the next proposal. The slice records which actions occurred and what each player observed following the game rule. At decision s , the acting player i knows the public history h_s , its own preferences v_i , and its private answers r_s^i . We write this view as $I_s^i = (h_s, v_i, r_s^i)$, an *information set* that identifies the game states the player cannot distinguish (Kuhn, 1953). The observed sequence locates the decisions to be labeled. For each such decision, the oracle also evaluates actions that were not taken and their possible continuations; it does not give the acting player access to the partners' hidden preferences. A formal slice definition appears in Appendix A. Following the Bayesian representation of private types (Harsanyi, 1967), the oracle solves the finite tree over possible preferences and actions under the common prior μ_0 . A *policy* $\pi_i(a \mid I)$ assigns a probability to each legal action $a \in \mathcal{L}(I)$ given what player i can know. Let Π_i contain all such policies, including plans for later decisions after each possible observation; π_{-i} denotes the other players' policies. Holding π_{-i} fixed, a *best response* maximizes player i 's expected terminal utility over these complete plans. Starting from uniform legal actions, we update every player against the same previous policy profile:

$$\text{BR}_i(\pi_{-i}) \in \arg \max_{\tilde{\pi}_i \in \Pi_i} \mathbb{E}_{\mu_0, (\tilde{\pi}_i, \pi_{-i})} [U_i(C_T; v_i)],$$

$$\pi_i^{(0)}(a \mid I) = |\mathcal{L}(I)|^{-1}, \quad \pi_i^{(k+1)} = \text{BR}_i(\pi_{-i}^{(k)}). \quad (3)$$

We retain a fixed profile π^* only if it passes a no-profitable-unilateral-deviation check, following the stability criterion of Nash (1950): in every initial private-information class, no player can improve its expected utility by changing its complete plan while the others keep theirs. At ACCEPT/REJECT responses, expected utility to the other players breaks ties in the responder's own utility; remaining ties are mixed uniformly. The reference policy also turns interaction into an inference target. Let $b_s^{i,*}(v_{-i}) = \Pr_{\mu_0, \pi^*}(v_{-i} \mid I_s^i)$ be player i 's *posterior*, its probability distribution over the others' hidden preferences. If partner j autonomously chooses a_s^j , Bayes' rule gives

$$b_{s+1}^{i,*}(v_{-i}) \propto b_s^{i,*}(v_{-i}) \pi_s^*(a_s^j \mid I_s^j(v_i, v_{-i})), \quad j \neq i. \quad (4)$$

An action is evidence to the extent that the reference policy makes it more likely under one candidate preference assignment than another. For planning, we instead ask what would happen if i chose each legal action a now:

$$Q_i^*(I_s^i, a) = \mathbb{E}_{\mu_0, \pi^*} [U_i(C_T; v_i) \mid I_s^i, \text{do}(a_s^i = a)]. \quad (5)$$

Table 1: Base-model diagnosis on 16 partner-generated histories, one per held-out structure. Regret is in terminal-utility units. Paired results use the 12 cases with valid actions under both supplied judgments.

Measurement	Result
Exact inference judgment	0/16
Optimal action with oracle judgment	1/16
Mean regret with oracle judgment	0.844
Optimal native history-to-action choice	1/15
Action changed by judgment replacement	11/12
Mean paired change Δ	-0.042

The forced action is followed by the reference policy for the rest of the game. Across a longer slice, each player’s posterior is updated from the observations available to that player and informs its later decisions. Appendix A details the computation and validity checks; Section 4.1 describes slice construction and sampling.

3 DIAGNOSE

We ask whether Qwen3-4B-Instruct-2507 (Qwen Team, 2025) can infer a partner’s preference from interaction and use that inference to choose a valuable action. A poor final decision could arise because the model misreads its partner, because it chooses poorly even given a correct reading, or because both steps fail. We therefore examine the two capabilities at the same decision, and replace the model’s explicit inference answer to test whether its choice depends on that answer.

Inferring from partner behavior. We evaluate 16 held-out two-player structures, eight with binary goals and eight with linear goals. Each case contains a partner offer chosen under the fixed reference policy, followed by the focal player’s response. The partner’s offer is evidence about its private preference: the oracle weights each candidate preference by the probability of that offer under the reference policy and updates the prior accordingly. The focal player’s own response changes the current commitments. The oracle reports a qualitative judgment J_c^* which means the preferences still possible and, when one has a sufficiently clear posterior lead, the favored preference. It also computes the expected terminal value $V_c(a) = Q_i^*(I_c, a)$ of every legal action under the fixed reference continuation. For every retained case, a certificate verifies that all posteriors compatible with J_c^* have the same optimal-action set. Thus omitted posterior magnitudes cannot explain a planning error when the correct judgment is supplied. The label rule and certificate are described in Appendix B.

Intervening on the judgment. For each case c , we first ask the model to infer a judgment $\hat{J}_c$ from the visible history. We then present the current decision in two fresh planning calls: one supplies $\hat{J}_c$, and the other supplies the oracle judgment J_c^* . Both calls omit the behavioral history and have everything else unchanged. A separate call asks the model to choose directly from the history. We measure decision quality by oracle regret and the paired change in value:

$$\text{Reg}_c(a) = \max_{a'} V_c(a') - V_c(a), \quad \Delta_c = V_c(\hat{a}(J_c^*)) - V_c(\hat{a}(\hat{J}_c)). \quad (6)$$

Here a positive Δ_c means that supplying the correct judgment improves the model’s chosen action. Both actions in the comparison are scored under the same oracle posterior.

Two failures and their interface. Table 1 reports the results on the 16 histories. The model never produces the exact qualitative inference judgment. Even when given the oracle judgment, it chooses an optimal action in only 1/16 cases (mean regret 0.844). Direct choice from the full history is optimal in 1/15 cases with a valid action. These observations identify failures in both evaluated interfaces under the fixed reference policy. Replacing the judgment changes the model’s action in 11/12 valid paired comparisons, so the explicit inference input affects its choice. The replacement improves value in only one case and has a mean paired change of -0.042. Thus a correct judgment does not reliably repair this model’s planning on the selected histories. The intervention concerns a judgment passed between calls; it makes no claim about separable internal modules.

An illustrative interaction. Consider a linear-goal case (`linear-12231`) in which Blair’s preference for ORCHARD is hidden. Orchard scores the average completion of Alex’s Cedar and Blair’s Cedar commitments. Blair also wants GARDEN, which depends on Alex’s Cedar and Blair’s Maple. Blair’s known preference for HARBOR and aversion to WORKSHOP cancel because those goals have identical requirements; it is neutral toward the remaining goal. Blair voluntarily offers to commit its own Maple without asking Alex to commit anything, and Alex accepts. Under the fixed reference continuation, that offer gives Blair expected terminal utility 0.5 for each possible Orchard preference. An offer that also asks Alex to commit Cedar or Maple would instead give Blair utilities 2, 1, and 0.5 when Blair respectively WANTS, is NEUTRAL toward, or AVOIDs Orchard. Thus the observed offer is suboptimal for WANT and NEUTRAL, but one of three tied optimal offers for AVOID. Its reference-policy likelihoods are consequently $(0, 0, 1/3)$; updating the uniform prior leaves only AVOID possible.

Alex now has the final proposal turn. Offering to add its own Maple while Blair retains its existing Maple raises Alex’s terminal utility from 0 to 1.5: three goals Alex wants each become half complete. Blair’s gain from HARBOR cancels its loss from WORKSHOP, so it accepts this offer on an own-utility tie. Investigating Blair’s Orchard preference instead consumes the last proposal turn and leaves Alex at 0. The base model reports all three preferences as possible and none as favored. With that judgment supplied to a fresh planning call, it investigates; when the same planner receives the oracle’s AVOID judgment, it makes the 1.5-utility offer. Asked to act directly from the history, the model instead proposes that both players add Cedar. An AVOID Blair rejects: accepting would lower its utility from 0.5 to 0, leaving Alex at 0. This case exhibits an inference error that the judgment intervention does repair, although such repair is uncommon across the suite.

4 EXPERIMENT

We evaluate whether training improves partner inference, action selection, and their composition in BENAC-I, and whether any gains transfer to other multi-agent tasks. We report the checkpoint and evaluation protocol with every result; update counts alone do not imply equal training-token budgets.

4.1 DATA AND TRAINING

We construct training items from slices of the finite BENAC-I game in Section 2.2. Instances vary goal dependencies, binary versus linear satisfaction, private preferences, commitment states, and interaction histories. We use the certified reference policy of Section 2.3 to obtain behavioral likelihoods for partner-state judgments and action values for legal decisions. We retain a supervised item only when its reference computation succeeds, and check generated transitions with the native game rules. Complete-game resets provide a separate source of interactive experience. The final dataset version, counts of distinct source structures and rendered training items, and the exact structural separation from each test suite are to be inserted here.

All arms initialize from Qwen3-4B-Instruct-2507. The *baseline* is the unchanged model. *SP* trains on complete self-play games using each player’s own terminal utility. *O* trains on history-to-action decision items, scored against the reference action values without a supplied partner judgment. *D* jointly trains *O*, partner inference, and planning given a correct qualitative partner judgment. In the planning task, the model sees the current decision state but not the behavioral history; the supplied judgment uses the same possible-set/favored interface as the inference task. These are alternative training objectives, not separate networks or an enforced inference-to-planning pipeline at inference time.

We use grouped policy optimization with eight responses per decision item; *SP* groups complete-game rollouts by reset and player. Training uses BF16, a 4096-token sequence limit, at most 1024 generated tokens per call, sampling temperature 1, and a KL penalty toward the frozen base model. The intended response-token budget is 6,553,600 per arm, with actual consumed tokens and checkpoint selection reported alongside each result. *D* collects four candidate groups from each of the history-to-action, inference, and planning tasks per update. Other optimization settings, including each arm’s learning-rate schedule and realized training dose, will be taken from the archived run configurations.

Table 2: Post-training diagnosis on 16 partner-generated histories, one per held-out structure. Planning regret uses the correct inference judgment. Regret and repair are means over valid outputs; their coverage may differ. Blank cells await a verified run under this protocol. BP-99 is a historical inference/planning-only reference, not one of the four main arms. Valid planning/native/repair counts are Q0 16/15/12, D-72 13/16/9, D-148 14/16/11, and BP-99 11/14/6.

Model	Inference exact	Planning regret	Native regret	Repair Δ_{repair}
Baseline (Q0)	0%	0.844	0.933	-0.042
SP	—	—	—	—
O	—	—	—	—
D-72	0%	0.769	0.750	+0.111
D-148	0%	0.821	0.469	0.000
BP-99 (historical)	0%	0.773	0.679	0.000

Table 3: Historical fixed-partner adversarial games. Utility is conditional on normal completion; completion counts are out of 24 per split. The final common-protocol O/D comparison remains to be filled.

Model	ID complete	ID utility	OOD complete	OOD utility
Baseline (Q0)	22/24	0.288	22/24	0.068
SP-69	20/24	0.525	18/24	-0.019
O	—	—	—	—
D	—	—	—	—
BP-139 (historical)	19/24	0.254	21/24	0.333

4.2 IN-DOMAIN DIAGNOSIS AND ADVERSARIAL GAMES

After training, we apply the partner-history, decision-sufficient diagnostic of Section 3 to each arm. We report exact semantic inference judgments, planning regret when the correct judgment is supplied, native history-to-action regret, and the paired judgment-repair effect. Regret is lower-is-better; repair is computed only where both paired actions are valid. The three deterministic repetitions of a structure are not independent test cases.

We also test full BENAC-I games with one focal model and two frozen baseline partners. Initial worlds are partitioned into seen dependency structures with new preference assignments (ID) and structurally held-out dependencies (OOD); focal seats are rotated. We report completed episodes and focal terminal utility conditional on completion separately, because an invalid or truncated game has no terminal utility. The currently archived comparison below is an earlier 48-episode, one-sample-per-scenario-seat evaluation and does not cover the selected O and D checkpoints.

4.3 TRANSFER TO CALBENCH AND SHAPEFACTORY

CalBench (Zou et al., 2026) asks privately informed agents to coordinate calendar meetings while managing individual scheduling costs. Our sequential transfer suite runs four homogeneous agents through three meetings in each of 24 frozen games. We measure full-stream completion and the number of meetings jointly scheduled; cost is interpreted only for successfully completed games. This small-model adaptation is distinct from the original benchmark aggregate score and from our fixed-partner BENAC-I test.

ShapeFactory in CollabSim (Chen et al., 2026) tests coordination through production, trade, and fulfillment of private orders. Our native three-agent setting uses three product specializations and one non-specialty order per agent. The same checkpoint controls all seats across two supply-cycle directions and private versus shared-dashboard visibility, for four games per model. We report fulfilled agent orders out of 12, games in which all three agents fulfill their orders, and final balance. The current v3 protocol retains these four ShapeFactory conditions; verified v3 model results are pending.

The transfer table is descriptive: the archived SP CalBench and ShapeFactory results use different checkpoints, and ShapeFactory’s four conditions are too few to support a broad generalization claim. The final comparison will use the same checkpoint, serving profile, output budget, and protocol for each model, and add per-condition outcomes and final balances.

Table 4: Available transfer observations. CalBench uses the 24-game sequential suite (three meetings per game). ShapeFactory values are from the earlier native v2 execution of the same four three-agent conditions; v3 results will replace these historical values. Checkpoints and training doses differ.

Model	CalBench complete	Meetings	ShapeFactory orders	All fulfill
Baseline (Q0)	5/24	37/72	1/12	0/4
SP-19 / SP-41	4/24	38/72	2/12	0/4
O-99	8/24	38/72	2/12	0/4
D-72	6/24	40/72	–	–
BP-99 (historical)	10/24	44/72	1/12	0/4

5 DISCUSSION

The game that we modified makes the connection between inference and planning measurable through explicit information sets and verified action values. We want to discuss three thought.

Richer partner states. Our diagnosis concentrates uncertainty on a partner’s preferences. In less constrained interactions, an agent must also infer what its partner has observed, what it believes others know, and which behavioral rule it follows. The same refusal, for example, could express an unfavorable preference or uncertainty about information unavailable to the partner. Recent work on multi-agent theory of mind examines private knowledge, adaptive reasoning about another agent’s mental state, and hypotheses about partner strategies (YS et al., 2026; Mu et al., 2026; Cross et al., 2025). These are extensions of *inference* as defined here: each concerns a latent state needed to predict a partner’s response and plan one’s own action. Extending game slices to such states would test whether the separation between inference and planning remains useful when beliefs themselves have structure.

Strategic coverage. The appeal of game slices lies in control over which decisions become training examples. Complete-game self-play generates coherent trajectories with the learner’s current partners, but its training distribution is the set of nodes those policies actually reach. A rarely visited offer, rejection, or information-gathering decision may therefore receive little training signal even when it is strategically decisive. We instead construct and sample slices around specified interaction patterns, retaining cases for which the reference computation supplies verified inference and action-value targets. Work on subgame curricula likewise shows that changing the distribution of starting states can accelerate multi-agent learning (Chen et al., 2024). This offers a possible explanation for the efficiency of targeted slices, though our comparison does not isolate state selection from denser supervision. The complementary strengths suggest a concrete hybrid: use complete games to discover consequential failures, solve and train on slices around those decisions, then return to full interactions to test whether the learned behavior survives its wider context.

Robustness across partner policies. Our reference labels assume one selected partner policy, while self-play primarily exposes the learner to copies of its evolving policy. This matters for inference as well as action selection: an offer that is impossible under one reference policy may be ordinary for a partner with a different planning horizon, tie rule, or degree of noise. Evidence that self-play agents can learn conventions that fail with unfamiliar partners, and that diverse training partners can improve cross-partner coordination motivates testing policy variation directly (Hu et al., 2020; Strouse et al., 2021). One direction is to train against a population of partner policies and generate slice labels under each policy, including controlled deviations from the reference solver. Belief targets could then marginalize over uncertainty about both preferences and behavioral policies, while evaluation holds out entire policy families. Such tests would distinguish strategic behaviors that transfer across partners from responses tailored to the particular oracle used here.

6 RELATED WORKS

Classical multi-agent decision making couples inference about other agents with downstream action selection, from opponent models and interactive beliefs (Carmel & Markovitch, 1996; Gmytrasiewicz & Doshi, 2005; Albrecht & Ramamoorthy, 2013) to belief-space coordination, inference

from actions, and opponent-aware planning (Macindoe et al., 2012; Wu et al., 2021; Foerster et al., 2019; Huang et al., 2024). For LLM agents, explicit mental-state or behavioral representations support collaboration, adaptation, trust tracking, and planning (Li et al., 2023; Cross et al., 2025; Jha et al., 2026; Mu et al., 2026; Zhou et al., 2026; Feng et al., 2026; Goel et al., 2026; Hwang et al., 2026). However, accurate mental-state prediction need not yield useful behavior (Riemer et al., 2025; Sobotka et al., 2026).

Recent MAS benchmarks expose strategic failures under private information, mixed incentives, and imperfect communication. Commitment-based negotiation, self-negotiated contracts, and multi-party conquest model consequential agreements and coalition dynamics (Benac et al., 2026; Wyse et al., 2026; O’Neill et al., 2026); CalBench, SOTOPIA-TOM, and SNEAK make information disclosure and privacy strategically consequential (Zou et al., 2026; YS et al., 2026; Avsian & Heck, 2026). Broader evaluations cover collaboration, controlled interaction, long-horizon commerce, and communication barriers (Zhu et al., 2025; Chen et al., 2026; Li et al., 2026b; Xuan et al., 2026). Diagnostic work further finds failures in distributed information pooling, distributed computation, dynamic grounding, and belief-sensitive negotiation (Li et al., 2026a; Zhang et al., 2026b; Yao et al., 2026; Zhang et al., 2026a; Agashe et al., 2025); these complement broad failure taxonomies and accounts of long-lived peer interaction (Cemri et al., 2025; Anthropic, 2026).

Game-based training has been used to induce transferable strategic and social reasoning through self-play, multi-game reinforcement learning, and social reasoning supervision (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b; Hua et al., 2026; He et al., 2026a; Wu et al., 2026). Cross-environment cooperation and Other-Play study generalization across tasks and partners (Jha et al., 2025; Hu et al., 2020). Curriculum work selects terminal-proximal or high-learning-value subgames (West et al., 2020; Chen et al., 2024), while LLM curricula can also induce undesirable strategic abstractions (Madmoun & Lahlou, 2026). Similarity-based inference has also been proposed as a route to cooperation among foundation-model agents (Meulemans et al., 2026).

7 CONCLUSION

AI USE STATEMENT

We used generative AI tools to assist with literature discovery, reference formatting, and drafting and editing portions of this manuscript. All AI-assisted text and bibliographic metadata were reviewed by the authors. The authors take responsibility for the final content of the paper, including its claims, experimental artifacts, and citations.

ETHICS STATEMENT

This work studies LLM-agent interaction in a synthetic, simulated negotiation environment and does not involve human participants or personal data. The environment includes mixed incentives and strategic information disclosure, which can surface behaviors such as manipulation or unhelpful information sharing. We use these behaviors only for controlled evaluation and training of coordination capabilities. We document the game rules, oracle policies, and limitations so that the resulting methods are not interpreted as deployment policies for real-world autonomous agents.

REPRODUCIBILITY STATEMENT

The main text and appendix specify the game rules, information structure, oracle belief filtering, and reference planning procedure. The appendix reports the diagnostic positions, prompts, evaluated model version, decoding configuration, aggregation procedure, random seed, and provenance information used for the reported results. These details are intended to make the diagnostic suite and its measurements reproducible.

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A GAME SLICES AND ORACLE SOLVER DETAILS

Multi-decision slices and information. Let $\mathcal{W} = \prod_{i \in \mathcal{N}} \Theta_i$ be the finite set of joint preference assignments, with prior μ_0 from Eq. 1. For a segment spanning decisions t through $u - 1$, a teacher-side record is

$$\sigma_{t:u} = (\mathcal{E}, v, h_t, (a_s, o_{s+1}, m_s)_{s=t}^{u-1}), \quad t < u, \quad (7)$$

where $v \in \mathcal{W}$ is the sampled joint preference assignment, h_t is the starting public history, a_s is the action actually taken, and $o_{s+1} = (o_{s+1}^j)_{j \in \mathcal{N}}$ records what each player observes afterward. The marker m_s distinguishes an autonomous action from an externally imposed one. The sampled world determines private investigation answers and terminal payoffs, whereas the actions are generated by the acting players or specified by an external starting context; v alone does not determine the action sequence. Decision indices count both proposals and responses, so $u = t + 1$ gives a single-decision slice.

The public history h_s determines commitments, the pending offer, the proposer schedule, and remaining investigation budgets. Player j additionally knows only v_j and its own private answers r_s^j . A slice can contain decisions by several players, each labeled from that player’s information set I_s^j ; the sampled world is never disclosed as part of this view. The observed sequence selects decision points to label, while the oracle retains all worlds in the prior’s support and all legal terminal continuations from each such point.

Finite-tree policy selection. A policy must assign the same action probabilities in any two worlds that a player cannot distinguish. We expand the public action tree for every $v \in \mathcal{W}$, retaining world-dependent private investigation answers. For a policy profile π , its terminal utility is $V_i^\pi(n, v) = U_i(C_T(n); v_i)$. At a nonterminal node n whose actor is j , the continuation values satisfy

$$V_i^\pi(n, v) = \sum_{a \in \mathcal{L}(n)} \pi_j(a \mid I^j(n, v)) V_i^\pi(n[a, v], v), \quad (8)$$

where $n[a, v]$ is the successor after action a in world v . The private answer to an investigation affects the investigator’s later information set, not the public information available to everyone else. Here $\mathcal{L}(n)$ denotes the legal actions at node n .

In Eq. 3, BR_i is a best response over i ’s *complete contingent policy*: it can revise later decisions after every possible observation, not merely the next action. We compute it by backward induction while keeping the other players’ policies fixed. Since the plan can depend on i ’s initial private information,

the maximization separates across root information classes with positive prior probability. At node n in world v , define the counterfactual reach weight

$$\lambda_{-i}(n, v) = \mu_0(v) \prod_{\substack{r \prec n: j_r \neq i \\ r \text{ autonomous}}} \pi_{j_r}(a_r \mid I_r^{j_r}(v)), \quad (9)$$

where $r \prec n$ ranges over earlier decisions on the path to n and j_r is the player acting at decision r . Conditioned on i 's information set, these weights determine the expected value of each candidate action. The product omits i 's own past choices, so an alternative plan remains evaluable even when it differs from i 's current plan; imposed moves contribute no behavioral likelihood. At a zero-reach information set, we use the root prior restricted to i 's own type and private investigation answers. We first maximize expected own terminal utility. Only at an ACCEPT/REJECT response do own-utility ties favor the action with greater expected total utility to others. Residual ties receive equal probability.

The updates for all players read the same previous profile and are applied simultaneously. A fixed profile is accepted only if, for every player and root private-information class,

$$\max_{\pi_i \in \Pi_i} \mathbb{E}_{(\pi_i, \pi_{-i}^*)}[U_i(C_T; v_i) \mid I_{\text{root}}^i] - \mathbb{E}_{\pi^*}[U_i(C_T; v_i) \mid I_{\text{root}}^i] \leq \varepsilon, \quad (10)$$

where Π_i contains legal information-respecting contingent policies and ε is the numerical tolerance. The check evaluates a full unilateral deviation, rather than only a one-step alternative. We discard cycles, timeouts, and profiles that fail the check. The resulting π^* is a selected reference profile, with no claim of uniqueness or of describing arbitrary human or LLM partners.

Belief filtering and action values. Let $b_s^{i,*}(v_{-i})$ be the joint posterior over all other players' types at one decision in the slice. If player $j \neq i$ autonomously takes public action a_s^j , define $\ell_s^*(v_{-i}) = \pi_j^*(a_s^j \mid I_s^j(v_i, v_{-i}))$. The candidate world determines what j privately knows, including answers to its own investigations. The exact update is

$$b_{s+1}^{i,*}(v_{-i}) = \frac{b_s^{i,*}(v_{-i})\ell_s^*(v_{-i})}{\sum_{\tilde{v}_{-i}} b_s^{i,*}(\tilde{v}_{-i})\ell_s^*(\tilde{v}_{-i})}. \quad (11)$$

An action by i or an imposed move changes the game state but contributes no likelihood about another player's type. A privately received investigation answer instead removes worlds inconsistent with that answer. We exclude observed actions whose total likelihood is zero. For every legal action at a selected decision, we force that action once and then follow the fixed reference profile:

$$Q_i^*(I_s^i, a) = \sum_{v_{-i}} b_s^{i,*}(v_{-i}) V_i^{\pi^*}(n_s[a, (v_i, v_{-i})], (v_i, v_{-i})). \quad (12)$$

For a queried partner-goal preference v_{jg} , the inference target is the positive-support set of $p_{jg}(z) = \sum_{v_{-i}: v_{jg}=z} b_s^{i,*}(v_{-i})$. The favored value is its mode only when that value exceeds the runner-up by more than 0.1; otherwise it is UNDETERMINED. Planning actions are evaluated using Q_i^* , including ties among optimal actions. The student sees neither the exact posterior nor action values. We use the same selected policy throughout a slice, while each player's posterior incorporates only its own observations. The stronger guarantee that a qualitative judgment alone determines the best action is certified separately for the selected partner-history diagnostic in Appendix B.

B PARTNER-HISTORY DIAGNOSTIC PROTOCOL

Selected histories and labels. This analysis uses 16 held-out two-player structures, eight with binary goals and eight with linear goals. Each contributes one history containing a reference-policy-consistent partner offer and a subsequent focal-player response. The selected histories have positive probability under the fixed reference policy, change the partner-preference posterior by at least 0.15 in total variation from the prior, and change the exact optimal-action set relative to the prior. Model outputs were not used for selection. The exact-posterior gap between the best and best non-optimal action ranges from 0.5 to 1.0 terminal-utility units (mean 0.781).

The queried uncertainty is one partner-goal preference in $\{\text{WANT}, \text{NEUTRAL}, \text{AVOID}\}$. For a visible partner choice a_s , the oracle multiplies each candidate preference's prior weight by the fixed

reference policy’s probability of a_s at that history, then normalizes. The focal player’s own response updates the physical state but does not supply another partner-action likelihood. The deployed inference interface returns the complete positive-support subset. It marks a value as favored when its posterior mass exceeds the runner-up by more than 0.1, and otherwise returns UNDETERMINED; it does not return probabilities. Both the model and correct judgments use identical serialization. The inference prompt identifies the partner’s offer as an observed choice. Condition names, case identifiers, correctness claims, posterior probabilities, and payoff values are absent from model prompts. Exact posteriors remain in the scorer and are used with the per-world payoff table to compute Eq. 6.

The correct semantic judgment is not a complete posterior, so selection explicitly certifies that the omitted magnitudes are decision-irrelevant. For each case, we express its support and favored-margin semantics as a finite union of rational convex polytopes and enumerate every vertex exactly. We also include zero-mass and threshold boundaries, obtaining a conservative superset of the strict semantic region. All vertices must induce the same exact optimal-action set. Since expected action value is affine in the posterior, this proves invariance over every compatible posterior; all 16 reported cases pass. Under this fixed policy, all 16 judgments have {AVOID} as their support. The suite therefore does not measure inference across all preference labels. In particular, the zero likelihood of an observed choice under some candidate preferences is a consequence of the selected policy, not an assertion that real partners never deviate from that policy.

Calls and intervention. For each case we obtain: (i) a native semantic inference judgment; (ii) a fresh planning call supplied with that judgment; (iii) a fresh planning call supplied with J^* ; and (iv) a native history-to-action call with no explicit judgment. The two controlled planning calls include the current binding state, remaining turns, investigation quota, known preferences, legal actions, and fixed partner policy, but exclude the behavioral chronology. Their tools, sampling parameters, and request seed are identical; only the serialized judgment may differ. If the model judgment is invalid, its composition cell is missing rather than replaced by a default action. On eight cases, a fifth call supplies inference with the reference planner’s qualitative likelihood ordering while still withholding all numeric probabilities.

Selection and scoring use the fixed teacher only. Changing a judgment never recomputes the partner continuation policy, and per-world payoffs respect the focal player’s information set rather than optimizing a different continuation separately in each hidden world. The paired quantity Δ_c is therefore the effect of changing the semantic input to the same model planner. It can have either sign and is not combined with planning regret as an additive error decomposition. Condition-specific regret means may have different valid subsets. On identical doubly valid cases the implementation checks $\Delta_c = \text{Reg}_c(\hat{a}(\hat{J}_c)) - \text{Reg}_c(\hat{a}(J_c^*))$ row by row and after aggregation.

Execution and aggregation. We evaluate the unchanged Qwen3-4B-Instruct-2507 checkpoint through vLLM using greedy decoding ($T = 0$, $\text{top-}p = 1$, $\text{top-}k = -1$), a 4,096-token completion limit, and three replicas. The frozen archive contains 96 returned rows; this analysis uses the 48 rows from partner-generated histories, representing 16 distinct decisions repeated three times. The replicas agree for every case, so they check execution determinism rather than provide independent samples. The run records its checkpoint, suite manifest, and runtime hashes. Every reported semantic certificate passes; the paired-scoring identity has zero row failures.

We collapse identical replicas and weight the 16 structures equally. Inference judgments and correct-judgment planning actions are valid in all 16 cases. Native history-to-action planning is valid in 15 cases. Both controlled planning calls are valid in 12 cases, which define the paired repair mean. Among these, replacing the judgment changes 11 actions; utility rises in one case, is unchanged in nine, and falls in two. On this identical paired subset, model-judgment regret is 0.833, correct-judgment regret is 0.875, and their difference equals mean repair -0.042 up to rounding. Correct-judgment regret over all 16 cases is 0.844 and uses a different subset. Qualitative reference-planner assistance changes no exact inference answer in the eight cases to which it is applied.

The frozen suite stores the model-facing cases, teacher-only certificates, selection record, and both clean controlled planning and native end-to-end requests. Hashes in the suite manifest detect changes to frozen cases and label dependencies before inference.